GCSE Mathematics - Foundation Tier

Paper 2 (Calculator) Practice - Set 5

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

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Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use pi = 3.142.

1.

Write 24% as a decimal.

(Total for Question 1 is 1 marks)

2.

A jacket costs £55. It is reduced by 20% in a sale. Work out the sale price of the jacket.

(Total for Question 2 is 2 marks)

3.

(a) Convert 4.2 metres to centimetres. (1)

(b) Convert 1.5 kilograms to grams. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 60 : 48 in its simplest form.

(Total for Question 4 is 2 marks)

5.

The line graph shows the temperature, in deg C, at a weather station every two hours during one day.

[Diagram: line graph - x-axis time (06:00 to 18:00 in 2-hour steps), y-axis temperature in deg C. Readings: 06:00 = 8, 08:00 = 12, 10:00 = 15, 12:00 = 19, 14:00 = 21, 16:00 = 17, 18:00 = 13.]

(a) Write down the maximum temperature shown on the graph. (1)

(b) Work out the increase in temperature from 06:00 to 14:00. (1)

(c) Between which two times did the temperature drop the most? (1)

(Total for Question 5 is 3 marks)

6.

Factorise fully $6x + 9$.

(Total for Question 6 is 2 marks)

7.

A walker covers 12 km in 2 hours 30 minutes.

Work out the average walking speed in km/h.

(Total for Question 7 is 3 marks)

8.

A student got the following marks in 6 maths tests:

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72   68   85   76   90   83
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Work out the mean test mark.

(Total for Question 8 is 3 marks)

9.

£600 is shared between four employees A, B, C and D in the ratio 1 : 2 : 3 : 4.

Work out how much employee C receives.

(Total for Question 9 is 3 marks)

10.

Solve $5(2x - 1) = 35$.

You must show all your working.

(Total for Question 10 is 3 marks)

11.

A garden has the shape shown - a rectangle 12 m long and 7 m wide with a square of side 3 m removed from one corner.

[Diagram: compound shape: 12 m by 7 m rectangle with a 3 m by 3 m square removed from one corner.]

Work out the perimeter of the garden. Give your answer in metres.

(Total for Question 11 is 3 marks)

12.

A pie chart shows the favourite type of book for 144 students.

[Diagram: pie chart with sectors - Fiction: 150 deg, Non-fiction: 90 deg, Biography: 75 deg, Comic: 45 deg.]

(a) How many students chose Non-fiction? (2)

(b) What percentage of students chose Comic? Give your answer to 1 decimal place. (2)

(Total for Question 12 is 4 marks)

13.

The cost C (in pounds) of a taxi journey is given by

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C = 3.50 + 1.20m
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where m is the distance in miles.

(a) Work out the cost of a journey of 8 miles. (2)

(b) Erin paid £15.50 for a journey. Work out the distance in miles. (2)

(Total for Question 13 is 4 marks)

14.

A laptop has a list price of £620. The shop adds VAT at 20% then offers a 10% discount on the VAT-inclusive price.

(a) Work out the price the customer pays. Show your working. (4)

(b) Express the price paid as a percentage of the list price. Give your answer to 1 decimal place. (1)

(Total for Question 14 is 5 marks)

15.

A length L is given as 12.7 cm correct to 1 decimal place.

(a) Write down the smallest possible value of L. (1)

(b) Write down the largest possible value of L (as the upper bound). (1)

A mass M is given as 350 g correct to the nearest 10 g.

(c) Write down the error interval for M in the form lower <= M < upper. (3)

(Total for Question 15 is 5 marks)

16.

A map uses a scale of 1 : 25 000.

(a) Two villages are 6 cm apart on the map. Work out the real distance between them in kilometres. (2)

(b) A river is 5 km long in real life. Work out its length on the map in centimetres. (2)

(Total for Question 16 is 4 marks)

17.

The table shows the age (in years) and the second-hand value (in £) of 8 used cars of the same model.

Age (yr)	1	2	3	4	5	6	7	8
Value (£)	14500	12800	10900	9200	7800	6400	5100	4200

(a) On the grid provided, plot these data as a scatter graph. (2)

(b) Describe the correlation. (1)

(c) Draw a line of best fit. (1)

(d) Use your line of best fit to estimate the value of a 4.5 year old car of this model. (1)

(Total for Question 17 is 5 marks)

18.

A swimming pool is in the shape of a rectangle 25 m long and 10 m wide, with a semicircle of diameter 10 m at one end.

[Diagram: plan view of a swimming pool - 25 m by 10 m rectangle with a semicircle of diameter 10 m attached to one of the 10 m ends.]

(a) Work out the total area of the pool. Give your answer to 1 decimal place. Use pi = 3.142. (4)

(b) The pool is 1.4 m deep throughout. Work out the volume in cubic metres. Give your answer to 1 decimal place. (1)

(Total for Question 18 is 5 marks)

19.

In a school survey, 120 students were asked how they spend most of their weekend.

Activity	Frequency
Sport	36
Studying	30
Gaming	24
Watching TV	18
Other	12

Draw an accurate pie chart to represent this information. Label each sector clearly.

(Total for Question 19 is 5 marks)

20.

A rectangle is twice as long as it is wide. Let the width be w cm.

(a) Write an expression in terms of w for the perimeter of the rectangle. (2)

The perimeter of the rectangle is 72 cm.

(b) Solve to find w, and hence the dimensions of the rectangle. (3)

(Total for Question 20 is 5 marks)

21.

A theatre has 800 seats. On Friday night 85% of the seats were sold. On Saturday night 92% of the seats were sold.

(a) Work out the number of seats sold on Friday night. (2)

(b) Work out the number of seats sold on Saturday night. (2)

(c) Work out the percentage increase in seats sold from Friday to Saturday. Give your answer to 1 decimal place. (2)

(Total for Question 21 is 6 marks)

22.

40 children chose one of three after-school clubs.

	Drama	Art	Coding	Total
Boys	___	___	9	20
Girls	6	8	___	20
Total	___	14	___	40

(a) Complete the two-way table. (3)

(b) A child is chosen at random. Work out the probability that the child is a boy who chose Coding. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 2 (Calculator) - Set 5 | 80 marks

Question 1

- 0.24 - B1

Total: 1 marks

Question 2

- Multiplier 0.8 or 20% of 55 = 11 - M1
- £44 - A1

Total: 2 marks

Question 3

- (a) 420 cm - B1
- (b) 1500 g - B1

Total: 2 marks

Question 4

- Both parts divided by a common factor (e.g. 30 : 24 then 15 : 12) - M1
- 5 : 4 - A1

Total: 2 marks

Question 5

- (a) 21 (deg C) - B1
- (b) 21 - 8 = 13 (deg C) - B1
- (c) Between 14:00 and 16:00 (drop of 4 deg) - B1

Total: 3 marks

Question 6

- Common factor 3 identified - M1
- 3(2x + 3) - A1

Total: 2 marks

Question 7

- 2 h 30 min = 2.5 h - M1
- 12 / 2.5 - M1
- 4.8 (km/h) - A1

Total: 3 marks

Question 8

- Sum = 72 + 68 + 85 + 76 + 90 + 83 = 474 - M1
- 474 / 6 - M1
- 79 - A1

Total: 3 marks

Question 9

- Total parts = 10; one part = 600 / 10 = 60 - M1
- C = 3 parts = 3 x 60 - M1
- £180 - A1

Total: 3 marks

Question 10

- Divide by 5: 2x - 1 = 7 (or expand: 10x - 5 = 35) - M1
- 2x = 8 (or 10x = 40) - M1
- x = 4 - A1

Total: 3 marks

Question 11

- Outer perimeter 2(12) + 2(7) = 38 - M1
- Cut corner replaces 3 + 3 with 3 + 3 (same length) -> perimeter unchanged at 38 OR 3 + 3 + 3 added back - M1
- 38 (m) - A1

Total: 3 marks

Question 12

- (a) 90 / 360 x 144 - M1
- (a) 36 (students) - A1
- (b) 45 / 360 x 100 - M1
- (b) 12.5 (%) - A1

Total: 4 marks

Question 13

- (a) 3.50 + 1.20 x 8 - M1
- (a) £13.10 - A1
- (b) 15.50 - 3.50 = 12 then / 1.20 - M1
- (b) 10 miles - A1

Total: 4 marks

Question 14

- (a) Multiplier 1.2 for VAT or VAT amount 124 - M1
- (a) Price after VAT 620 x 1.2 = 744 - M1
- (a) 10% discount: 744 x 0.9 or 744 - 74.40 - M1
- (a) £669.60 - A1
- (b) 669.60 / 620 x 100 = 108.0 (%) - B1

Total: 5 marks

Question 15

- (a) 12.65 (cm) - B1
- (b) 12.75 (cm) - B1
- (c) 345 stated (lower) - M1
- (c) 355 stated (upper) - M1

- (c) 345 <= M < 355 - A1

Total: 5 marks

Question 16

- (a) 6 x 25 000 = 150 000 cm - M1
- (a) 150 000 / 100 / 1000 = 1.5 km - A1
- (b) 5 km = 500 000 cm; / 25 000 - M1
- (b) 20 cm - A1

Total: 4 marks

Question 17

- (a) 8 points correctly plotted - B2 (B1 if 6+ correct)
- (b) Negative correlation (strong) - B1
- (c) Sensible line of best fit - B1
- (d) Value estimated at 4.5 yr (accept £8200 - £8800) - B1

Total: 5 marks

Question 18

- (a) Rectangle area 25 x 10 = 250 - M1
- (a) Semicircle area (1/2) x pi x 5^2 (radius 5) - M1
- (a) 0.5 x 3.142 x 25 = 39.275 - M1
- (a) Total area 289.3 (m^2) (accept 289.2 - 289.4) - A1
- (b) Volume 289.3 x 1.4 = 405.0 (m^3) (accept 404.8 - 405.2) - B1

Total: 5 marks

Question 19

- 360 / 120 = 3 deg per student - M1
- At least 3 angles correct: Sport 108, Studying 90, Gaming 72, TV 54, Other 36 (deg) - M1
- All 5 angles correct - A1
- Pie chart drawn within 2 deg tolerance - B1
- All sectors labelled - B1

Total: 5 marks

Question 20

- (a) Length = 2w identified - M1
- (a) Perimeter = 2w + 2(2w) = 6w - A1
- (b) 6w = 72 - M1
- (b) w = 12 - A1
- (b) Width 12 cm, length 24 cm - B1

Total: 5 marks

Question 21

- (a) 0.85 x 800 - M1
- (a) 680 seats - A1
- (b) 0.92 x 800 - M1
- (b) 736 seats - A1

- (c) Increase 736 - 680 = 56; / 680 x 100 - M1
- (c) 8.2 (%) - A1

Total: 6 marks

Question 22

- (a) Boys row complete: Drama + Art + 9 = 20 so Drama + Art = 11; with Art column total 14 and girls art 8, boys art = 6; so boys drama = 5 - M1
- (a) Girls coding = 20 - 6 - 8 = 6 - M1
- (a) Drama total = 5 + 6 = 11; Coding total = 9 + 6 = 15 - A1
- (b) Boys & Coding = 9, total = 40 - M1
- (b) 9/40 - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 2 (Calculator) - Set 5 | Detailed explanations

Question 1

To convert a percentage to a decimal, divide by 100.
24% = 24 / 100 = 0.24.

Question 2

Sale price = 80% of £55.
0.8 x 55 = 44.
Answer: £44.
Or: 20% of £55 = £11, so sale price = £55 - £11 = £44.

Question 3

(a) 1 m = 100 cm, so 4.2 m = 4.2 x 100 = 420 cm.
(b) 1 kg = 1000 g, so 1.5 kg = 1.5 x 1000 = 1500 g.

Question 4

HCF of 60 and 48. Factors of 60: ..., 12, 60. Factors of 48: ..., 12, 48. HCF = 12.
60 / 12 = 5, 48 / 12 = 4.
Answer: 5 : 4.

Question 5

(a) Read the highest point: 21 deg C (at 14:00).
(b) Temperature rose from 8 deg at 06:00 to 21 deg at 14:00, an increase of 13 deg C.
(c) Drops occur after 14:00. 14:00 -> 16:00: 21 -> 17 (drop of 4). 16:00 -> 18:00: 17 -> 13 (drop of 4). They are equal in size but the steepest single 2-hour interval is from 14:00 to 16:00 (the first big drop).

Question 6

Find a common factor of 6x and 9. HCF = 3.
6x + 9 = 3(2x) + 3(3) = 3(2x + 3).

Question 7

2 h 30 min = 2.5 hours.
Speed = distance / time = 12 / 2.5 = 4.8.
Answer: 4.8 km/h.

Question 8

Sum = 72 + 68 + 85 + 76 + 90 + 83 = 474.

Mean = sum / count = 474 / 6 = 79.

Question 9

Total parts = 1 + 2 + 3 + 4 = 10.

One part = £600 / 10 = £60.

C receives 3 parts = 3 x 60 = £180.

Question 10

5(2x - 1) = 35. Divide by 5: 2x - 1 = 7.

Add 1: 2x = 8. Divide by 2: x = 4.

Check: 5(2(4) - 1) = 5 x 7 = 35.

Question 11

The "cut corner" replaces a 3 m + 3 m bit of the outer rectangle with another 3 m + 3 m route around the square. The perimeter is therefore unchanged.

Perimeter of outer rectangle = 2(12) + 2(7) = 24 + 14 = 38 m.

Answer: 38 m.

Question 12

(a) Non-fiction: (90 / 360) x 144 = 0.25 x 144 = 36 students.

(b) Comic: (45 / 360) x 100 = 12.5%.

Question 13

(a) C = 3.50 + 1.20 x 8 = 3.50 + 9.60 = 13.10. Answer: £13.10.

(b) 15.50 = 3.50 + 1.20m. So 1.20m = 12, giving m = 10 miles.

Question 14

(a) Add 20% VAT: 620 x 1.2 = £744.

Apply 10% discount to the £744: 744 x 0.9 = £669.60.

Customer pays £669.60.

(b) £669.60 as a % of £620 = (669.60 / 620) x 100 = 108.0%.

Question 15

(a) The smallest value that rounds to 12.7 is 12.65 (lower bound).

(b) The upper bound is 12.75 (any value at or above this would round to 12.8).

(c) Nearest 10 means half-of-10 = 5 either side.

Lower: 350 - 5 = 345. Upper: 350 + 5 = 355.

Error interval: 345 <= M < 355.

Question 16

(a) 6 cm on map = 6 x 25 000 = 150 000 cm in real life.

Convert to km: 150 000 cm = 1500 m = 1.5 km.

(b) 5 km = 5000 m = 500 000 cm. On map: 500 000 / 25 000 = 20 cm.

Question 17

(a) Plot 8 points.

(b) Negative correlation - older cars have lower values.

(c) Best fit drawn through the points.

(d) At 4.5 years the line gives roughly £8200 - £8800.

Question 18

(a) Rectangle area = 25 x 10 = 250 m^2.

Semicircle radius = 5 m. Area = (1/2) x pi x 5^2 = 0.5 x 3.142 x 25 = 39.275 m^2.

Total area = 250 + 39.275 = 289.275, i.e. 289.3 m^2 (1 dp).

(b) Volume = area x depth = 289.275 x 1.4 = 404.985, i.e. 405.0 m^3.

Question 19

Angle per student = 360 / 120 = 3 deg.

Sport: 36 x 3 = 108 deg.

Studying: 30 x 3 = 90 deg.

Gaming: 24 x 3 = 72 deg.

Watching TV: 18 x 3 = 54 deg.

Other: 12 x 3 = 36 deg.

Check: 108 + 90 + 72 + 54 + 36 = 360 deg.

Question 20

(a) Width = w, length = 2w.

Perimeter = 2 x w + 2 x 2w = 2w + 4w = 6w.

(b) 6w = 72, so w = 12.

Width 12 cm, length 24 cm. Check the perimeter: 2(12) + 2(24) = 24 + 48 = 72 cm.

Question 21

(a) Friday: 0.85 x 800 = 680 seats.

(b) Saturday: 0.92 x 800 = 736 seats.

(c) Increase = 736 - 680 = 56.

% increase = (56 / 680) x 100 = 8.235...%, i.e. 8.2% to 1 dp.

Question 22

(a) Boys row total = 20. Boys Coding = 9, so Boys Drama + Boys Art = 11.
Art column total = 14, Girls Art = 8, so Boys Art = 14 - 8 = 6. Boys Drama = 11 - 6 = 5.
Girls row total = 20. Girls Drama = 6, Girls Art = 8, so Girls Coding = 20 - 6 - 8 = 6.
Column totals: Drama 5+6=11, Art 14, Coding 9+6=15. Grand total: 11+14+15 = 40.
(b) Boys and Coding = 9. Probability = 9 / 40.